

Seokju Lee

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EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) <i>Ph.D. Candidate - Mechanical Engineering; Advisor: Prof. Kyung-Soo Kim</i> <i>Mechatronics, Systems and Control (MSC) Lab</i>	Daejeon, South Korea Mar 2025 – Present
Korea Advanced Institute of Science and Technology (KAIST) <i>M.S. - Mechanical Engineering; Advisor: Prof. Kyung-Soo Kim</i> <i>Mechatronics, Systems and Control (MSC) Lab</i> <i>M.S. Thesis: Slip-Compensated Legged Robot State Estimation Using Latent Space Attention Mechanisms</i>	Daejeon, South Korea Mar 2023 – Feb 2025
Aalto University <i>Exchange Student</i>	Espoo, Finland Jan 2022 – Feb 2022
Ulsan National Institute of Science and Technology (UNIST) <i>B.S. - Electrical Engineering; GPA: 4.07/4.3; Major GPA: 4.13/4.3</i> <i>Summa Cum Laude, Three-Year Early Graduation, Graduate Representative</i>	Ulsan, South Korea Mar 2020 – Feb 2023

HONORS AND AWARDS

- UNIST Best Students Awards (Minister of Science and ICT Award)
- Presidential Science Scholarship
- Daejeon Design Thinking Hackathon 1st Award (Minister of Environment Award)

PUBLICATIONS

* Equal contribution † Equal advising

Journal Articles

- [1] **Seokju Lee** and Kyung-Soo Kim. Attention-Based Neural-Augmented Kalman Filter for Legged Robot State Estimation. *IEEE Robotics and Automation Letters (RA-L)*, 2026. [\[link\]](#)
- [2] Yunji Jung, **Seokju Lee**, Tair Djanibekov, Jong Chul Ye, and Hyunjung Shim. Text Optimization with Latent Inversion for Non-Rigid Editing. *Pattern Recognition Letters*, 2025. [\[link\]](#)
- [3] Hyun-Bin Kim, **Seokju Lee**, Byeong-Il Ham, Keun Ha Choi, and Kyung-Soo Kim. Temperature Compensation Method of Six-Axis Force/Torque Sensor Using Gated Recurrent Unit. *IEEE Sensors Journal*, 2025. [\[link\]](#)

Conference Papers

- [1] **Seokju Lee***, Jeongtae Lee*, Jeonghyeok Lim, Jeonguk Kang, Byungwook Lee, Seungho Han, Keun Ha Choi, Dongil Park†, and Kyung-Soo Kim†. ADP: Adversarial Dynamics Priors for Physically Grounded Humanoid Locomotion. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2027. (Under Review)
- [2] **Seokju Lee**, Jeonghyeok Lim, Jeongtae Lee, Jeonguk Kang, Seungho Han, Dongil Park, and Kyung-Soo Kim. Learning Humanoid Locomotion via Disentangled State Reconstruction. In *International Conference on Control, Automation and Systems (ICCAS)*, 2026. (Under Review)
- [3] Byungwook Lee, Jeonguk Kang, **Seokju Lee**, Kwang-Woo Jeon, Tae-Hwan Kim, Hyun-Joon Chung, Man bok Hong, Jangsik Park, and Kyung-Soo Kim. Optimal Load Distribution Strategy via Knee-Hip Coordinated Control for Heavy Load Lifting Assistance in Wearable Robots. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026. (Accepted)
- [4] Antyanta Bangunharcana, **Seokju Lee**, and Seungho Han. Bridging Reactive Simulation and Log-Playback for Safer Autonomous Driving with GRPO. *Asian Control Conference (ASCC)*, 2026. (Accepted)
- [5] **Seokju Lee**, Hyun-Bin Kim, and Kyung-Soo Kim. Legged Robot State Estimation Using Invariant Neural-Augmented Kalman Filter with a Neural Compensator. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. pages 15445-15452. IEEE, 2025. [\[link\]](#)
- [6] **Seokju Lee**, Seunghun Jeon, and Jemin Hwangbo. Learning Legged Mobile Manipulation Using Reinforcement Learning. In *International Conference on Robot Intelligence Technology and Applications (RiTA)*. pages 310-317. Cham: Springer International Publishing, 2022. [\[link\]](#)

Preprints

- [1] Seungho Han, **Seokju Lee**, and Jeonguk Kang. RAY-TOLD: Ray-Based Latent Dynamics for Dense Dynamic Obstacle Avoidance with TDMPC. *arXiv preprint arXiv:2604.27450*, 2026. [\[link\]](#)

INVITED TALKS

Institute of Control, Robotics and Systems (ICROS) 2026 <i>Selected Recent Papers, AI in Control Session</i> <i>"Attention-Based Neural-Augmented Kalman Filter for Legged Robot State Estimation"</i>	Daegu, South Korea Jul 1, 2026
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EXPERIENCE

Korea Advanced Institute of Science and Technology (KAIST) <i>Robotics and Artificial Intelligence Lab, Research Intern</i>	Daejeon, South Korea <i>Jun 2022 – Aug 2022</i>
◦ Learning legged mobile manipulation using reinforcement learning (Advisor: Prof. Jemin Hwangbo)	
<i>Teaching Assistant</i>	<i>Mar 2025 – Dec 2025</i>
◦ Linear System Control, Multidisciplinary Capstone Design, Mechanism Design	
Ulsan National Institute of Science and Technology (UNIST) <i>Robotics and Mobility Lab, Research Intern</i>	Ulsan, South Korea <i>Dec 2021 – Nov 2022</i>
◦ Develop the Autonomous Platform to deliver service using sensor fusing (Advisor: Prof. Jeong hwan Jeon)	
<i>AI Graduate School Creative Self-Challenge Contest</i>	<i>May 2021 – Feb 2022</i>
◦ Team Leader; Lead 3D modeling and control research for quadrotor	
<i>Teaching Assistant</i>	<i>Sep 2021 – Nov 2022</i>
◦ Calculus I, Calculus II	

PROJECTS

Learning Blind Humanoid Locomotion: (Work with Korea Institute of Machinery & Materials (KIMM)) Reinforcement-learning-based humanoid locomotion in complex terrains. (June 2025 – Present)
Autonomous Flying Robot based on Structural Battery: (Work with Agency for Defense Development (ADD)) Learning-based control and autonomous soaring strategies for an ultra-high-efficiency surveillance flying robot enabled by structural battery technology. (Aug. 2023 – Dec. 2025)
Interworking Control for a Lower Limb Exoskeleton: (Work with Korea Institute of Robotics and Technology Convergence (KIRO)) Position-based knee-hip interworking control for lifting-assistance exoskeletons, verified through human lifting experiments and sEMG-based muscle activity analysis. (Jan. 2024 – Dec. 2024)

SKILLS SUMMARY

Programming Languages:	Python, C, C++, MATLAB, ROS
Simulators:	Isaac Gym, Isaac Sim, Raisim, Mujoco, Gazebo

PROFESSIONAL SERVICES

Conference Reviewer:	ICRA, IROS
Journal Reviewer:	IEEE Transactions on Robotics (T-RO), Pattern Recognition, Neurocomputing, Measurement, Artificial Intelligence Review